Tidal potential measured on a coastal–inland optical-clock baseline

Introduction

Measurements of Earth's gravity field reveal how water, air and the solid Earth redistribute mass. Ground gravimetry measures gravitational acceleration and its spatial gradients, while precise levelling combined with gravity observations determines potential differences.[1,2] Satellite missions have tracked large-scale changes in water storage and the climate system.[3,4] More demanding goals include following short-lived oceanic and atmospheric mass changes through their effects on the potential difference between distant sites, and separating contributions that overlap in time.[5] Tides illustrate the latter problem. The Moon's and Sun's tidal fields act together with deformation of the solid Earth and redistribution of ocean water. Atmospheric pressure tides also force the ocean,[6] while polar motion produces a slower pole tide.[7] Solid-Earth tidal potentials can reach amplitudes of several $\mathrm{m}^2/\mathrm{s}^2$ at individual sites, while smaller contributions place greater demands on sensitivity and temporal coverage.[8]

Optical clocks offer a further way to measure potential differences through their relative ticking rates. Near Earth's surface, a fractional frequency change of 10^{-18} corresponds to about 1 cm of equivalent height.[9] Experiments have resolved gravitational redshifts on laboratory scales and measured static potential differences between distant sites.[2,10,11,12] Optical clocks now reach fractional systematic uncertainties near 10^{-19}, and separate lattice-clock experiments have achieved instabilities of a few $10^{-17}/\sqrt{\tau/\mathrm{s}}$, where τ is averaging time.[13,14] Optical fibres connect clocks across laboratories,[15,16,17] while long-distance free-space experiments are testing the transfer capability needed for future ground-to-space links.[18,19] These networks support optical timekeeping and a future redefinition of the second, as well as comparisons of gravitational potential.[9,20] Pole tides and non-tidal mass changes in the atmosphere, oceans and continental water can matter at the 10^{-18} fractional-frequency level.[8] Resolving such time-varying signals depends on clock and link noise over the duration of the change. Observation gaps further limit the phase coverage and can leave the frequency reference level uncertain between runs.

Here we detect tidal variations in the potential difference between a strontium clock in coastal Shanghai and a ytterbium clock in inland Wuhan. The clocks are separated by 692 km and connected by a 1,350-km fibre link.[21] The combined solid-Earth and ocean-loading prediction gives an M_2 fractional-frequency amplitude of 4.78×10^{-18}, equivalent to about 4.4 cm of height.[7,22] We use 545.9 hours of observations spread over 353 days. The astronomical phase is known at each observation time, even when the record is interrupted. We therefore allow each observing segment its own frequency offset and combine its measured variations with those from other dates. The changing relative phase of the lunar and solar tides permits joint estimation of the M_2+N_2 and neighbouring S_2+K_2 responses, with phases and relative amplitudes fixed within each group.

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